

Qatar-1b

R-2735

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_260621 · created 2026-09-02T00:43:07+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2461212.7034 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.018
Transit depth (Rp/Rs)^2	2.03 +/- 0.52 [%]
Orbital Inclination (inc)	83.54 +/- 1.7
Airmass coefficient 1 (a1)	0.999 +/- 0.011
Airmass coefficient 2 (a2)	-0.001 +/- 0.01
Scatter in the residuals of the lightcurve fit is	1.13 %
Best Comparison Star	#2 - [257, 471]
Optimal Method	PSF photometry
Transit Duration (day)	0.063 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.142 $\pm$ 0.018 vs 0.1463 (deviation 0.24 σ)
Duration fit vs published	0.063 d vs 0.0692 d (deviation 0.47 σ)
Mid-time O-C	-7.8 min
Depth SNR	7.9
Residual scatter	1.13 %

Light curve

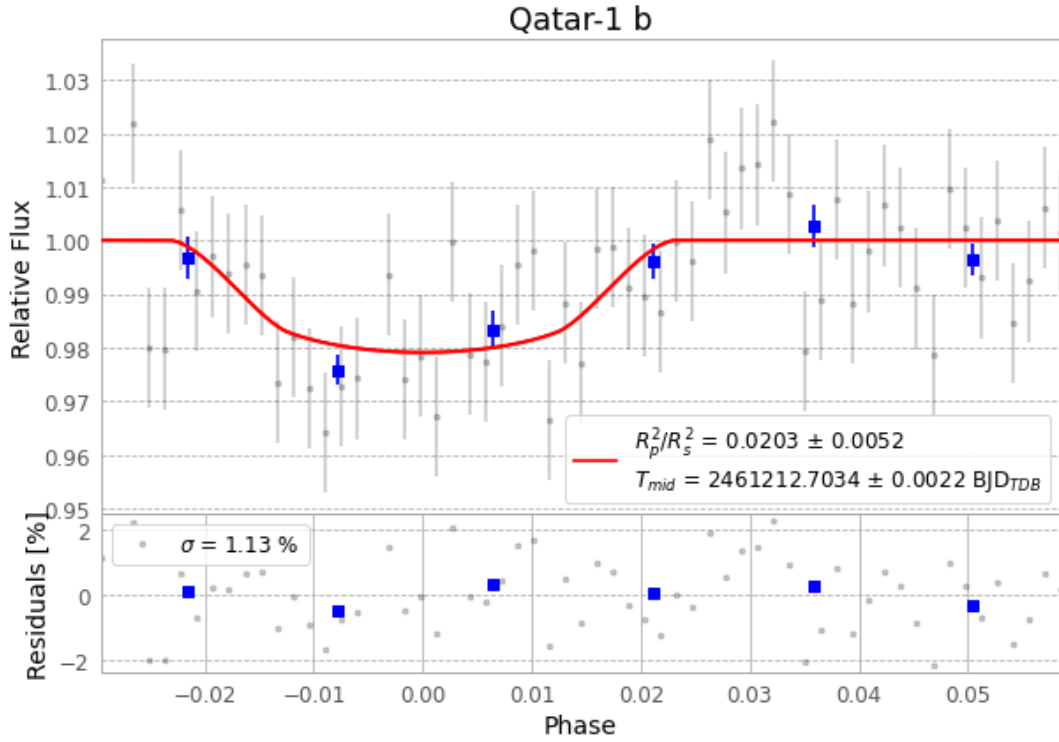


Figure 1 — FinalLightCurve_Qatar-1 b_2026-06-20.png (SHA-256 fea1f4add76daafc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [247, 304], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9c41776844a40487...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), Qatar-1260621035115.FITS → Qatar-1260621065116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-260621.log (9ab2f10cde3d...), FinalLightCurve_Qatar-1 b_2026-06-20.png (fea1f4add76d...), FinalLightCurve_Qatar-1 b_2026-06-20.csv (113c6b4f438f...)

Compute resources

Wall time 131.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 00:43Z.